

The Hidden Side of Oil Spills: Are They Limited to What We See?

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ABSTRACT

Each oil spill has a unique impact depending on its hydrocarbon composition and the environmental conditions in which it occurs. Prolonged exposure to high concentrations of hydrocarbons can lead to various health impairments, including immunosuppression, skin lesions, cancers, and adrenal disorders. Polycyclic aromatic hydrocarbons (PAHs) are among the most harmful compounds, being classified as priority hazardous substances. Natural biodegradation processes require time, which is why dispersants are often applied to oil slicks. These substances emulsify the oil, promoting faster degradation. However, the combined impact of dispersants and hydrocarbons on ecosystems is more harmful than hydrocarbons alone. Indeed, dispersants increase the bioavailability of PAHs, leading to higher mortality rates in exposed species.

Introduction

Maritime transport, like any form of transport, is not without risk. However, in this case, the damage caused by an accident is not confined to the area of the incident. Oil slicks are carried by currents and wind; as a result, damage can be observed hundreds of kilometers away from the spill site.

Since the development of industry, human impacts on the world's seas and oceans have been numerous. These range from visible pollution, such as drifting fishing nets and macroplastics thrown into the sea, to invisible pollution like the discharge of contaminants or microplastics resulting from the degradation of plastics. Concerns are many and varied. Since the late 1960s, a new concern has emerged: "oil spills." The term "oil spill" was coined by a journalist during the wreck of the Torrey Canyon in 1967, referring to the flow of an oil slick into coastal areas. Crude oil, which belongs to hydrocarbons, is a fossil fuel found in porous rocks in the form of droplets at

CONTEXT

Maritime transport, like any form of transport, is not without risk. However, in this case, the damage caused by an accident is not confined to the area of the incident. Oil slicks are carried by currents and wind; as a result, damage can be observed hundreds of kilometers away from the spill site.

The solutions implemented after a spill are often highly questionable from the perspective of surrounding ecosystems. The dispersion of hydrocarbons in the water column does not reduce their toxicity and may even promote faster spread.

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varying depths. Once extracted through drilling, it is sent to refineries where it is processed to produce gasoline or various plastic products, for example [1]

Each oil spill has a different impact, occurring in a unique context that depends on the quantity and properties of the hydrocarbons released as well as the rate of the spill. Moreover, the characteristics of the spill area (coastal zones, open sea), the weather (winds, currents), and the type of ecosystem affected all influence the harmful effects of oil spills [2].

Some impacts are visible (water contamination, trapped birds, mass fish deaths, etc.), but most are invisible, such as the migration of hydrocarbons through the water column. Each spill, with its own context and composition, is often difficult to manage using a single standardized protocol. Every year, new techniques are considered in order to minimize their negative effects on the impacted biotopes.

What Is an Oil Spill

Definition of the phenomenon

Although the term “oil spill” automatically implies a leak of hydrocarbons of human origin to the general public, this is not entirely accurate. In reality, an oil spill is defined as the release of hydrocarbons into the marine environment. This release can be natural, through seepage from storage areas, or anthropogenic, via cargo ships, offshore installations, or pipelines following accidents [3]. Unlike crude oil, the composition of hydrocarbons released during an accident is highly variable, depending on the refining processes they have undergone [4].

The *Erika* was an oil tanker carrying 31,000 tons of heavy fuel from Dunkirk (France) to Livorno (Italy). Following adverse weather conditions in December 1999, the ship broke in two, releasing between 7,000 and 10,000 tons during the sinking off the coast of Brittany. Since then, the term “oil spill” has been widely used, as more than 700 similar accidents have occurred in our seas and oceans since this event. However, this number, as significant as it is, does not account for natural hydrocarbon releases, which still represent a substantial percentage of aquatic discharges [4].

The direct effects and impacts of an oil spill are difficult to identify and quantify. However, a cause-and-effect correlation can be established when, after a spill, malformations appear, fertility rates decline, or certain marine populations (bivalves, gastropods, fish, etc.) decrease [5]. Nevertheless, proving this correlation remains challenging, as pollution from pesticides or heavy metals can produce identical symptoms [6].

Regarding marine fauna, it is strongly impacted during large anthropogenic hydrocarbon spills. Media coverage often shows images of emblematic animals coated in oil to illustrate these disasters. However, some species (such as salmon and cod) have been observed exhibiting avoidance behaviors, indicating a capacity to detect pollutants. These animals subsequently return to their habitats when pollutant concentrations are lower [6, 7].

Natural hydrocarbon release

During the natural formation of oil, hydrocarbon leaks occur through small faults in the ground [4]. These natural leaks release approximately 600,000 tons of hydrocarbons worldwide each year, which accounted for 45% of hydrocarbon releases in the 1990s (**Figure 1**) [4].

Source	Estimation moyenne (t)	Part (%)
Suintements naturels	600 000	47
Déversements opérationnels	306 000	24
Déversements à terre	145 000	11
Accidents du transport pétrolier	119 000	9
Apports atmosphériques	60 000	5
Extraction pétrolière	38 000	3
Total	1 268 000	100

Figure 1: Sources and volumes of hydrocarbons spilled yearly in the oceans from 1990 to 1999 [8] ; [9].

These natural seep areas are a very reliable indicator of the presence of an underground hydrocarbon source. That is why most hydrocarbon extraction stations are located near these natural sources. This co-location between natural seeps and extraction stations explains why these leaks were long ignored, or even attributed to offshore facilities. This highlights the difficulty of measuring and estimating the amount of hydrocarbons released by these leaks each year worldwide [10].

The difference between these natural seeps and oil spills caused by, for example, the sinking of a tanker, lies in the quantity of hydrocarbons released, the location of the seepage, and the duration of the discharge. Natural seeps are random and release only small amounts of hydrocarbons, making estimates particularly challenging. In contrast, when a tanker runs aground, tons of hydrocarbons are released in large quantities at a single location [8].

Classification of Different Types of Hydrocarbons

The behavior and toxicity of a hydrocarbon depend on its chemical composition. By definition, a hydrocarbon consists of carbon and hydrogen atoms. They can be classified into different groups based on the number of atoms and the types of bonds between them [11].

Saturated hydrocarbons have only single bonds and are called alkanes. They can be acyclic, with a linear or branched chain, or cyclic, such as naphthenes. Cycloalkanes have a cyclic structure but do not contain any aromatic sites [11].

Unsaturated hydrocarbons are divided into two groups: cyclic chains with one or more aromatic sites, and linear chains with one or more double bonds, also called alkenes. Hydrocarbons without aromatic rings are also called aliphatic hydrocarbons. Finally, aliphatic compounds with a cyclic chain but still without aromatic sites are called alicyclic hydrocarbons [11, 12] (**Figure 2**).

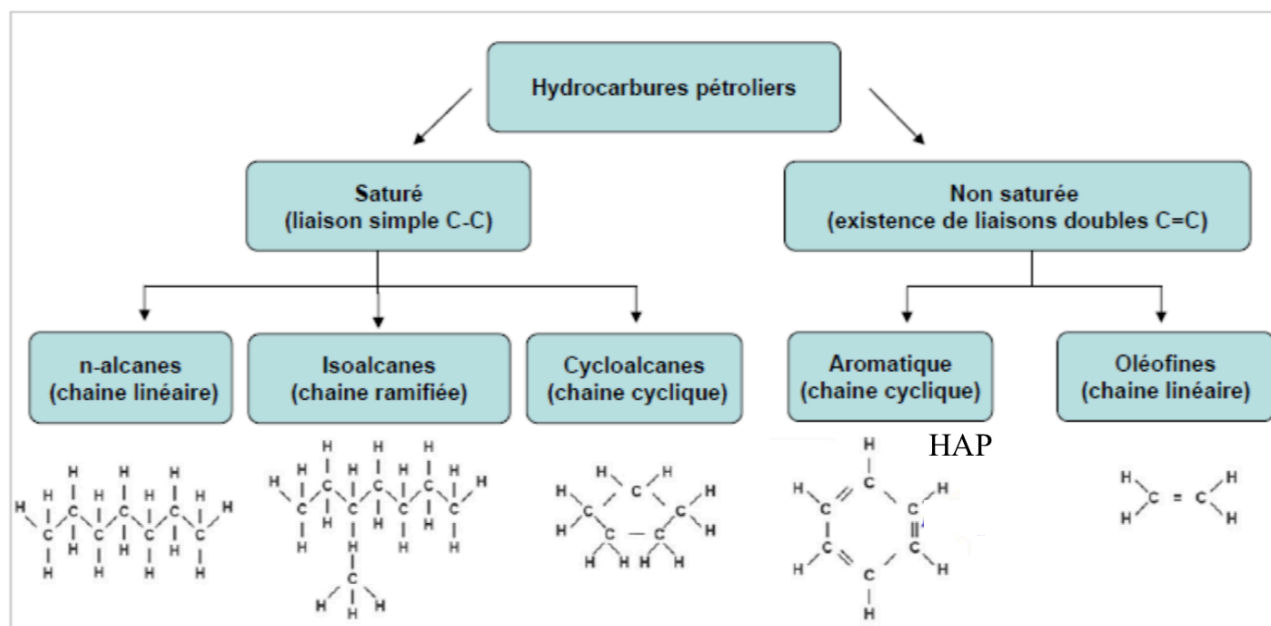


Figure 2: Basic molecular structure of major petroleum hydrocarbons [11].

The type of oil is determined by the dominance of a particular type of hydrocarbon: saturated (paraffinic type), unsaturated (mixed type), or aromatic (bituminous type). Refining allows crude oil to be processed into various kinds of petroleum products with specific concentrations by exploiting the boiling points of each hydrocarbon [12, 13]. Polycyclic aromatic hydrocarbons (PAHs) are the most harmful hydrocarbons and are considered priority hazardous substances under the Water Framework Directive (Article 16-2) [14].

Consequences of an Oil Spill Once Released

Pathways of Hydrocarbon Entry

It is important to understand that not all hydrocarbons are equally toxic. Aliphatic hydrocarbons and naphthenes (cyclic saturated hydrocarbons) are insoluble in seawater and therefore less likely to disperse or dissolve in the water column. In contrast, mono-aromatic compounds are insoluble in water but highly lipophilic and also volatile, meaning that a portion will be released into the atmosphere [12]. Finally, hydrocarbons can be ranked by toxicity, and their effects can be influenced by various abiotic factors such as wind and temperature [2].

The buoyancy and dispersion behavior of different hydrocarbons varies depending on their composition as well as abiotic factors such as wind, sunlight, and temperature [2]. Different

behaviors are observed: some molecules evaporate, others settle, and some float in the water column (**Figure 3**).

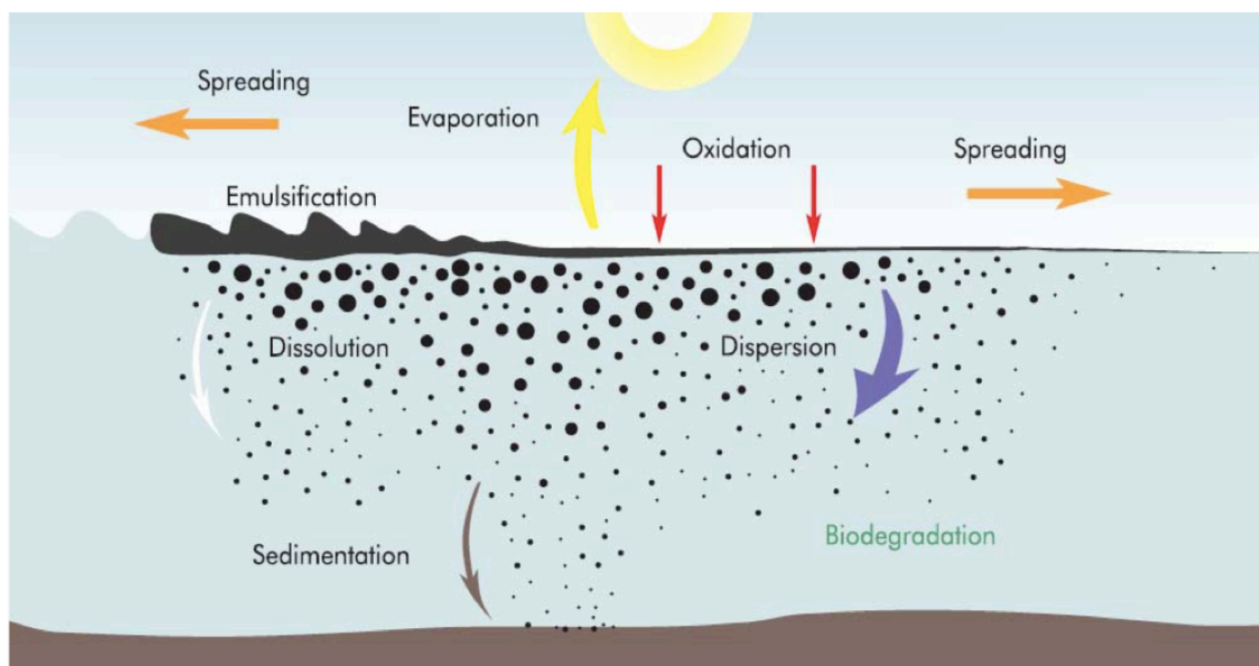


Figure 3: Behavior of hydrocarbons in the water [15].

Heavy fuel oil has slightly lower buoyancy than seawater, so it tends to sink toward the seafloor. However, slicks exposed to waves are subject to emulsification, which increases their buoyancy and allows them to travel longer distances [16]. Oil particles that settle on the seabed may take several years to dissipate to natural concentrations (between 0.2 and 2.0 $\mu\text{g/g}$) [17, 18]. Particles remaining on the surface can travel to the coasts, having a significant impact on seabird populations as well as on communities living along the shoreline. These phenomena have been observed repeatedly, for example during the *Exxon Valdez* spill in 1989, the *Erika* in 1999, and the *Prestige* in 2002 [16].

In contrast, oil spills containing light fuel have a smaller impact on birds but a much greater impact on plankton populations and marine animals [16]. Light and toxic molecules, such as polycyclic aromatic hydrocarbons (PAHs), are absorbed by organisms, causing irreversible damage (see “Impacts of an Oil Spill on Marine Fauna and Flora”). PAHs are highly lipophilic, meaning they can accumulate in animal fat [12].

Polycyclic aromatic hydrocarbons (PAHs)

Polycyclic aromatic hydrocarbons (PAHs) are actually formed by the fusion of multiple carbon rings and also contain alkyl groups [2]. Common PAHs include naphthalene, acenaphthene, fluorene, anthracene, fluoranthene, pyrene, and many others [19].

The higher the number of rings, the less volatile, soluble, and biodegradable they are, and the more hydrophobic and toxic they become [2]. They are naturally present in bitumen, tar, or oils released during oil spills and, as mentioned earlier, are considered the most harmful hydrocarbons.

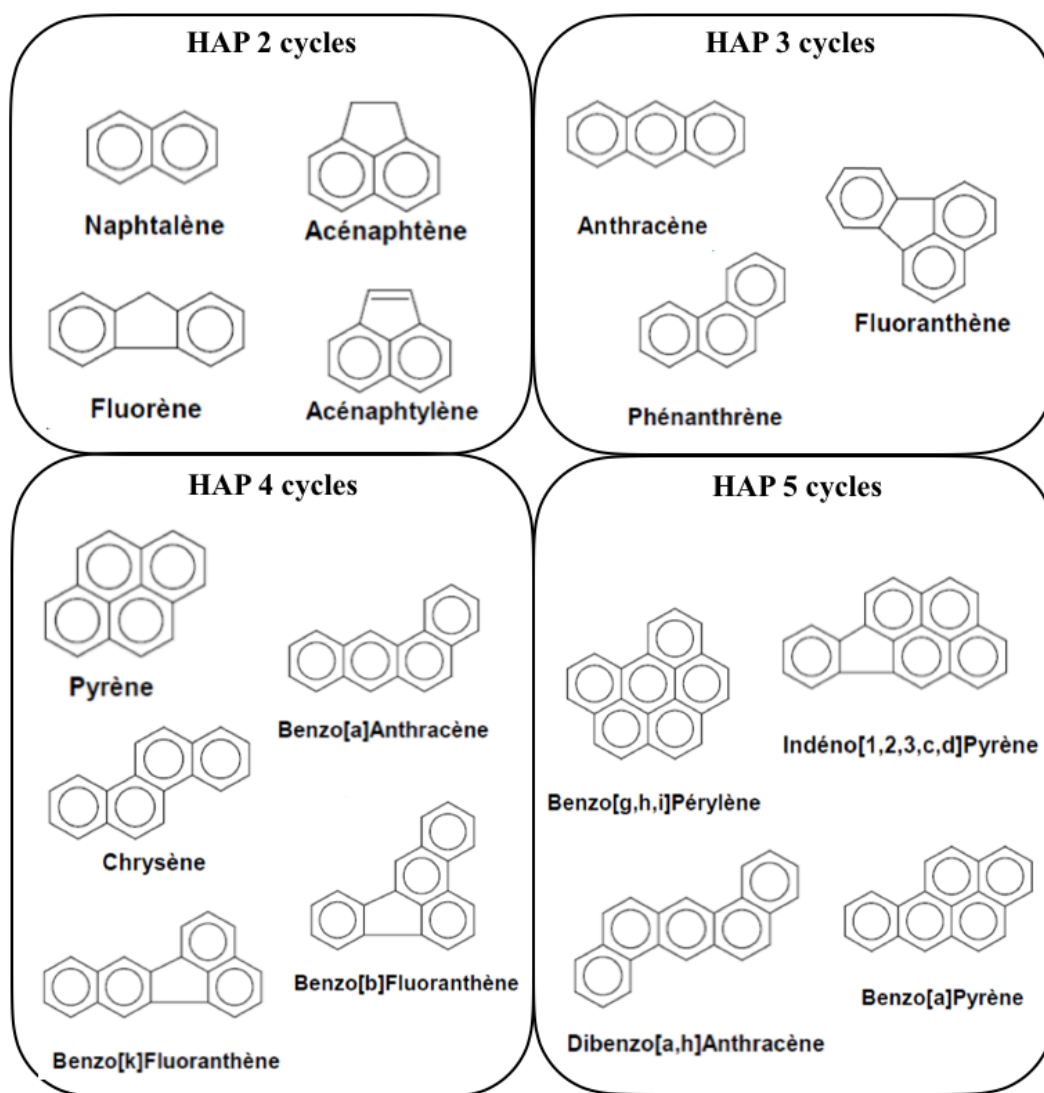


Figure 4: List of priority PAHs according to US-EPA recommendations. Inspired from [5]; [23].

Humans release them through the combustion of fossil fuels, asphalt, or during the cracking of crude oil for refining [20]. The US Environmental Protection Agency (US-EPA) and the European Environment Agency (EEA) have listed 16 polycyclic aromatic hydrocarbons as priority pollutants [21] (Figure 4). The United Nations Environment Programme (UNEP) has defined harmful PAH concentrations as those above 10 µg/g and harmless concentrations as those below this threshold [20]. The presence of PAHs at harmful concentrations can have various effects on fauna and flora due to their genotoxicity, mutagenicity, and carcinogenicity [5, 21, 22].

PAHs enter the ocean either directly through oil spills, maritime transport, or coastal discharges, or indirectly via atmospheric or riverine inputs. Atmospheric inputs are the primary source of PAHs in the marine environment, depositing as residues on the water surface. The pathway by which PAHs enter the water depends on their carbon composition [24, 25].

PAHs pose a major environmental threat due to their toxicity. Moreover, biological activity in surface water layers causes pollutant particles (PAHs and POPs) to sink toward the seabed. High-molecular-weight PAHs do not dissolve in water and tend to attach to other particles, which enhances their migration to the seafloor [25]. During their descent, they can be absorbed by organisms through gill filtration or via the food chain [25, 26].

Fate of an Oil Slick

Hydrocarbon Degradation

The biodegradation of hydrocarbons largely depends on the location of the oil spill. In a warm environment that has been previously exposed to hydrocarbons, resident organisms—particularly microbial communities—have adapted, allowing for faster degradation [27]. In contrast, hydrocarbons released in the open sea tend to degrade more slowly due to low temperatures, increased salinity, and reduced sunlight [2].

The degradation of an oil slick occurs in several stages. Volatile compounds evaporate, and sunlight has an oxidative effect on certain molecules, initiating their breakdown by making them more polar and thus more soluble in water [5]. Then, wind and waves transport part of the slick, causing it to drift. Over time, it gradually degrades: some components settle on the seafloor, while others either dissolve or remain suspended in the water column [2, 28] (**Figure 5**).

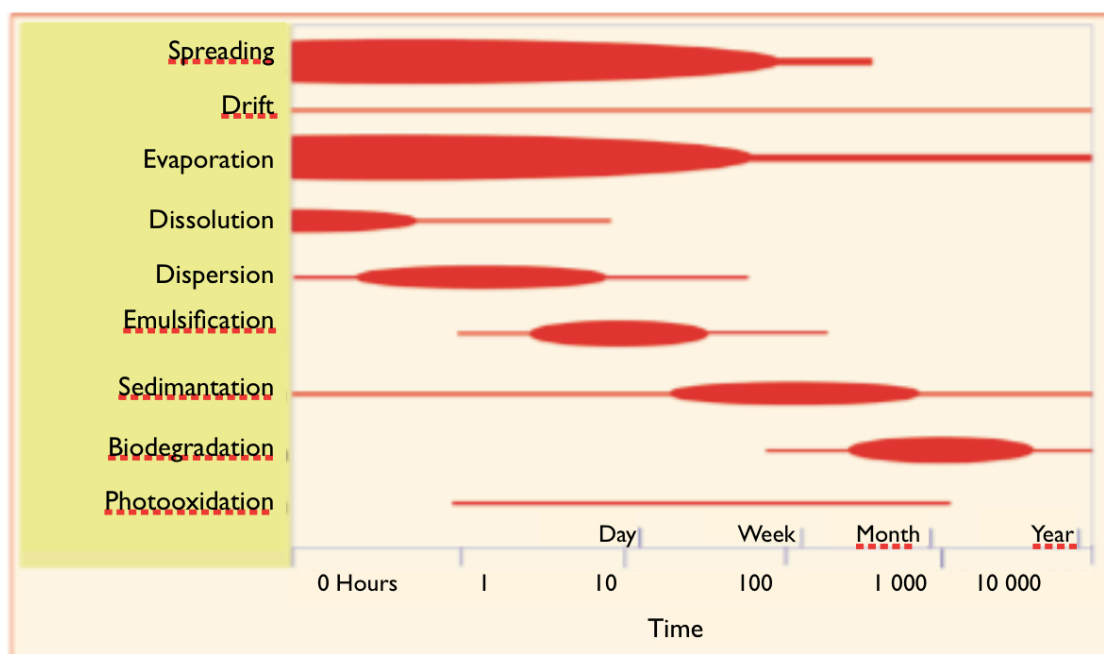


Figure 5: Various hydrocarbon removal processes without dispersants over time [27].

Some degradation processes, particularly oxidation reactions, generate intermediate compounds. Thus, during the breakdown of hydrocarbons, acids, esters, alcohols, phenols, and other compounds are produced and released into the environment [27].

PAHs, both with and without alkyl groups, represent a large portion of crude hydrocarbons. Moreover, their degradation is generally slower than that of other compounds, such as alkanes. The higher the degree of alkylation of PAHs and the more rings they contain, the slower their degradation [27].

Regarding alkanes, their oxidation begins with the addition of an oxygen atom (terminal oxidation), forming a primary alcohol, which is then converted into a carboxylic acid. This carboxylation reduces the initial hydrophobicity of alkanes, increasing their bioavailability, making

them more easily assimilated and more readily eliminated [5]. Similarly, mono-, di-, and polycyclic aromatic hydrocarbons are also degraded into a mixture of aliphatic aromatic acids in the aqueous phase [27].

Concerning biodegradation by marine organisms, it has been shown that some organisms are capable of degrading carboxylic acids, alkylated and non-alkylated PAHs, as well as polycyclic aromatic acids resulting from oil degradation [27]. Carboxylic acids are degraded via β -oxidation or excreted by microorganisms. However, this degradation generates new molecules, such as naphthalene derivatives for *Pseudomonas putida* CSV86, which degrades naphthalenes. Similarly, the fungus *Cunninghamella elegans* produces polycyclic aromatic acids while degrading PAHs. Finally, the polychaete worm *Nereis diversicolor* forms polycyclic aromatic acids while degrading alkylated PAHs [27].

The most common pathway for non-alkylated PAHs in bacteria is the dioxygenase pathway, where oxidation of an aromatic ring produces several byproducts. Alkylated PAHs can be degraded either via the dioxygenase pathway with initial aromatic ring attack or by hydroxylation of the alkyl side chains [29]. This hydroxylation leads to the formation of primary alcohols, which are then converted into aldehydes and subsequently into carboxylic acids [27].

Certain polychaetes are also known to degrade hydrocarbons through their digestive processes, transferring hydrocarbons into the soluble phase by producing surfactants in the gut, as observed in *Nereis* species. During passage through the worm's intestine, the solubility and dissolution capacity of hydrocarbons increase, making them more available for biodegradation [30, 31].

Hydrocarbon degradation after an oil spill can be assessed using the nC17/Pristane and nC18/Phytane ratios, which serve as biodegradation indicators. Hydrocarbon degradation is considered negligible as long as n-alkanes are still available as substrates. These ratios therefore allow monitoring of hydrocarbon biodegradation following a spill [27, 30, 33].

Regarding crude oil, more than 70% is rapidly degraded within the first month following release. This is facilitated by the degradation methods mentioned above, as well as by macrofauna (mollusks, crustaceans, annelids, etc.). Initially, these organisms bury hydrocarbons while supplying oxygen to typically anaerobic zones. Subsequently, particles are resuspended in the water column through seabed mixing, diluting hydrocarbons over a larger volume and lowering their concentration. However, this mechanism does not operate if hydrocarbons are located too deeply in the sediment. Nonetheless, the vast majority of hydrocarbons are released into the water column rather than stored in marine sediments. In addition, hydrocarbons on the seabed are degraded under anaerobic conditions, as well as through oxidation facilitated by mixing that returns them to aerobic zones [30].

It is important to note that the impact of an oil spill on nearby flora and fauna depends not only on the composition of the oil but also on the characteristics of the affected biotope and the ecosystem's biocoenosis. Not all hydrocarbon families are subject to natural degradation; some remain stagnant at the surface, suspended in the water column, or deposited on the seabed [30]. This is why dispersants are used during oil spills to break up the oil slick into smaller fractions.

Chemical dispersant

Chemical dispersants are a tool frequently used in oil cleanup operations, applied in about 18% of global oil spill incidents [33]. Their principle is simple: they break the oil into smaller particles so that the non-miscible droplets can dissolve into the water column. After the accident in the Gulf of Mexico, tons of Corexit 9500 (a chemical dispersant) were sprayed on the oil slick [34].

Natural dispersion occurs, as explained earlier, when waves move the oil slick, causing it to break apart and migrate into the water column. When a breaking wave passes over an oil slick at sea, the slick is temporarily broken into small droplets. Most oil droplets are large and quickly rise back to the surface, forming a new oil layer after the wave passes, while only the smallest droplets disperse into the water column. A chemical dispersant is added to the slick to accelerate this natural process and rapidly convert a larger portion of the oil into very small droplets [35] (**Figure 6**).

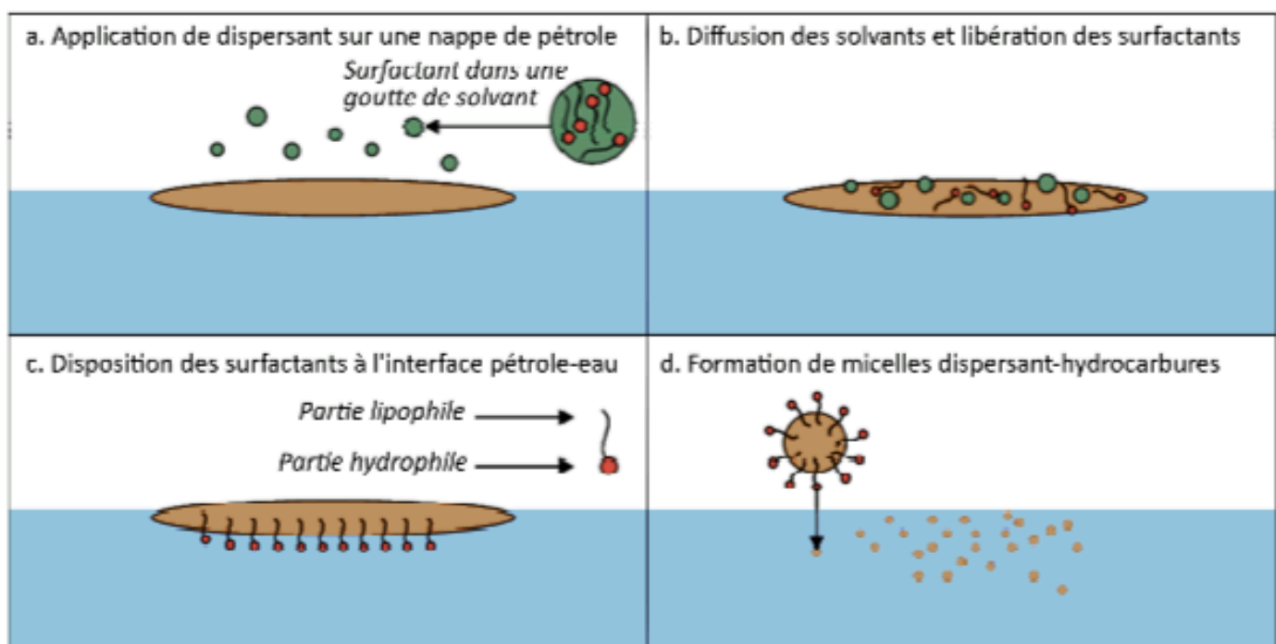


Figure 6: Schematic representation of the mode of action of dispersants [23].

Dispersant agents are composed of surfactants (surface-active agents) dissolved in one or more solvents. The surfactant molecules have hydrophilic “head” groups that associate with water, and lipophilic “tails” that associate with oil. These surfactants are combined with one or more aqueous or organic solvents and one or more stabilizing agents [36]. Oil droplets are thus surrounded and stabilized by these surface-active molecules, which promotes rapid dilution by water movement [37].

After the Deepwater Horizon (DWH) disaster, tons of the chemical dispersants Corexit 9500 and Corexit EC9527A were sprayed on the oil slick [34, 37]. According to the US EPA (2006), these products contain 10–30% detergent and 1–5% propylene glycol, which are considered non-hazardous substances. However, Corexit 9500 contains 10–30% petroleum distillates, and Corexit EC9527A contains 30–60% 2-butoxyethanol, which can be toxic [38].

Many of the first dispersants used in the 1970s and 1980s showed very high toxicity to marine organisms. However, laboratory data today indicate that modern dispersants (so-called

“third generation”) have relatively low toxicity to marine organisms. Since chemical dispersants increase the availability of oil in the water column, there is a risk that this could have harmful effects on organisms [39]. Efforts continue to develop new biodegradable dispersant formulas, such as tests NF. T. 90-346 and NF. T. 90-345 [23]. These have shown a positive environmental benefit when dispersants are used in certain coastal areas, such as mangroves [40, 41].

Numerous experimental studies have been conducted, using LC50 (Lethal Concentration killing 50% of exposed individuals) measurements to determine toxicity in terms of mortality following dispersant application [23]. A study at the School of Environmental Studies, Queen’s University in Canada, examined the effect of a dispersant on trout. Two groups of fish were exposed to PAHs: one combined with dispersants and the other to hydrocarbons alone. The study showed that using a dispersant increased fish exposure to PAHs present in crude oil [42].

Another study conducted by CNRS in France tested the effect of two chemical dispersants, produced by Total Fluides and Innospech, on PAH bioaccumulation and mortality. This study also concluded that fish exposed to oil with a chemical dispersant showed higher bio-concentration of PAHs compared to fish exposed to oil alone. They also observed higher mortality in fish exposed to chemical dispersants [12].

Other studies have been conducted on numerous species in coastal ecosystems, including fish [43], amphipods [44], cephalopods [45], echinoderms [46], and macroalgae [47]. All these experimental approaches showed that mortality increased following dispersant application.

Impacts of an Oil Spill on Marine Fauna and Flora

Impacts on Fauna

Pollution caused by drilling and exploration activities is difficult to prevent, first because of possible dispersion and second because of the inaccessibility of the environment. Two types of pollution are associated with these activities: chronic pollution and acute pollution [48].

Acute pollution can be illustrated by the Gulf of Mexico incident. It consists of oil discharges at sea, whether accidental spills during routine operations or larger leaks due to malfunctions on platforms [49]. Chronic pollution is more varied and can include simple noise pollution (from tankers or platforms) harmful to marine fauna, the discharge of production waste, or increased water turbidity [48].

Industry activities have numerous environmental repercussions. For example, atmospheric emissions from flaring are concerning because they affect air quality and greenhouse gas levels. Pollutant emissions from the industry include toxic substances such as benzene, smog precursors, acidic emissions, and greenhouse gases like methane and carbon dioxide [50].

Regarding the impacts of hydrocarbons on marine fauna and flora, the Deepwater Horizon (DWH) incident has been the main case study. This event, in addition to being catastrophic, is very recent, allowing for thorough study and monitoring of affected species [2].

Zooplankton, which consumes 70% of marine snow and occupies the lower levels of the food chain, is a major player in bioaccumulation across trophic levels [2]. Exposure of zooplankton to hydrocarbons has significant short-term effects. While some taxa decreased in density, others increased. For copepods, exposure to crude oil alone reduced egg production and hatching rates [37]. In contrast, some species appear less sensitive (such as scyphozoans and ctenophores) and can serve as vectors for contaminant transfer along the food web to predators such as sea turtles [18, 34]. Following the DWH accident, sea turtles experienced increased mortality and stranding rates, averaging five times higher according to NOAA data [51].

Following the same incident, invertebrates—particularly polychaetes—were completely decimated, and nematode diversity dropped to 58%. It took five years for meiofaunal communities to return to normal. This is because hydrocarbons cause disorientation, reduced activity, slower growth, and decreased burrowing rates, leaving organisms more exposed to predators. Reduced growth rates were directly correlated with PAH concentrations in marine sediments [39]. Larval stages were often observed to be more sensitive to hydrocarbons than adult stages [28, 52].

Crustaceans were also severely affected. After the sinking of the *Prestige* in Spain, macrobenthic community richness declined by 67%. Hydrocarbon deposits in sediments following a spill cause sediment hardening and reduce fauna abundance, both of which hinder crab recolonization after an oil spill. Additionally, some organisms, such as sand crabs, use their setae-covered antennae to capture suspended particles. However, certain polycyclic hydrocarbons destroy these setae, preventing these organisms from feeding [2].

Filter-feeding organisms, such as mussels, are among the first affected by an oil spill. Hydrocarbons filtered through the gills cause DNA damage [2]. Mollusks were severely impacted, being nearly entirely decimated during the DWH disaster. Some species continued to disappear even two years after the incident. Moreover, in the presence of dispersants, burial behavior was suppressed in the gastropod *Polinices conicus* [44].

Oil can contaminate nektonic organisms such as fish, marine mammals, and seabirds. It can also embed itself in sandy beaches and coastal ecosystems as black tar-like patches [28]. In this form, it can affect unique and threatened ecosystems, such as sandy beaches, estuaries, mangroves, and coral reefs [53].

In vertebrates as well as plants, the enzymes used to degrade PAHs are oxidases and reductases because, as in plants, PAHs induce the production of reactive oxygen species (ROS), which can cause cell death. In humans, the harmful effects of hydrocarbons are well established, causing various cancers including bladder, lung, kidney, pancreas, and others [2]. Therefore, it is reasonable to assume that, although effects vary among organisms, PAHs remain harmful to marine life.

The main symptoms observed in fish include skin lesions, reduced growth, decreased larval survival, and lower fecundity—all indicators of immunosuppression. Fish exposed to hydrocarbons are thus more susceptible to opportunistic pathogens [2]. Killifish embryos exposed to hydrocarbons at the DWH spill site exhibited multiple developmental anomalies, including failure to hatch, reduced vitality, and the presence of pericardial and yolk sac edema. Cardiomyocytes, the heart

muscle cells, are auto-excitabile—they can depolarize independently and transmit this depolarization to neighboring cells, enabling cardiac automaticity. Hydrocarbon exposure, however, can disrupt potassium and calcium ion regulation, interfering with membrane signal transduction and causing cardiac irregularities in fish [54]. The study by Milinkovitch et al. (2011) highlighted that dispersants are significantly more toxic than untreated, undispersed oil alone [12]. Moreover, increased mortality and higher PAH bio-concentration in fish muscle were observed, likely due to increased bioavailability of oil in droplet form. Jung et al. (2009) showed that in rockfish (*Sebastes schengeli*), exposure to dispersed oil decreased brain acetylcholinesterase activity, suggesting impaired neural function [55].

Marine mammals occupy the top of the marine food chain. As noted above, they ingest substantial amounts of hydrocarbons, which, like in humans, have detrimental health effects. Their defense system mainly comprises specialized cells and proteins capable of recognizing and combating harmful molecules (antigens). These antigens may originate from bacteria, viruses, or toxic agents. PAHs are among these antigens and can lead to dysfunction and a reduction in lymphocyte numbers, impairing immune response. This immune deficiency is called immunosuppression. Combined with reduced resources, it increases mortality rates among marine mammals following exposure to and ingestion of hydrocarbons [56]. Consequently, marine mammals become more susceptible to bacterial diseases such as primary bacterial pneumonia, contributing to whale deaths. Like humans, cetaceans may develop lung diseases and adrenal disorders, leading to adrenal toxicity, as observed in several dolphins exposed to oil [2]. Regarding dispersant impacts, some have been identified as cytotoxic and genotoxic to whale dermal fibroblasts [2, 57]. These cells produce extracellular matrix components (collagen and elastin fibers) that contribute to skin flexibility and strength. During wound healing, they generate growth factors and matrix materials to form repair tissue. In the presence of dispersants, healing tissue cannot form or persist, potentially causing fibrosis [2, 57].

During the DWH spill, shorebirds were the most severely affected tetrapods. Oil coating has a particularly prolonged effect on shorebirds, remaining on individuals for over six months after the incident. In addition to causing direct acute mortality, hydrocarbons also impair health and behavior, sometimes with near-lethal effects [53]. Habitat degradation due to hydrocarbon pollution deprives seabirds of prey and nesting sites. Oil spills can thus impact multiple seasons, causing severe consequences for various communities.

Corals are slow-growing and highly sensitive to environmental disturbances, as illustrated by coral bleaching due to global warming [58]. Hydrocarbons have a devastating effect on coral colonies, potentially affecting over 90% of corals at a spill site [14]. Exposure to hydrocarbons induces severe stress, causing polyp retraction, reduced proliferation, tissue loss, and tissue bleaching [2, 59]. During DWH, dispersant residues containing hydrocarbon droplets were found in deep-water corals 30 km from the spill site. Dispersants have far more destructive effects on corals than hydrocarbons alone [3]. Additionally, Epstein et al. (2000) reported altered swimming behavior in larvae of two coral species (*Heteroxenia fuscescense* and *Stylophora pistillata*) in the presence of dispersants, likely due to impaired nervous system functionality [60].

Impacts on Flora

The impacts of oil spills are felt from phytoplankton to mammals, including seabirds. Photosynthetic organisms form the base of the food chain, and oil particles are bioaccumulated and transferred through various trophic levels [2, 61]. Additionally, phytoplankton appears to be affected by hydrocarbons, which reduces chlorophyll content and leads to a decrease in biomass [61]. Dispersants have an even more devastating effect on phytoplankton communities than hydrocarbons alone [53].

Marine flora is extremely sensitive to physicochemical changes in its environment. During hydrocarbon pollution, plants are quickly affected, as was the case with seagrass beds during the DWH incident, where over 109 hectares were decimated. This vulnerability is due to hydrocarbons' ability to penetrate the vascular system and intercellular spaces, causing damage to membrane structures. This leads to apoptosis of damaged cells and, in the long term, the death of the plant. Furthermore, oils present in hydrocarbons clog stomata and intercellular spaces, reducing respiration and consequently photosynthesis, directly affecting root growth and overall plant survival [2].

Measures to Be Taken:

Regulation of Oil Transport and Offshore Platforms

Numerous rules govern the legislation of oceans and seas. Some are relatively old, such as the United Nations Convention on the Law of the Sea, known as the Montego Bay Convention, which dates back to 1982. It defines rules for free communication and navigation between countries, peaceful use of the seas, fair and efficient exploitation of resources, and protection of the marine environment [62]. Around the same time, an international treaty for the prevention of pollution from ships was introduced, known as the MARPOL (Marine Pollution) Convention. This treaty is the primary legal instrument concerning marine environmental protection [63].

In the mid-1990s, an international convention on oil pollution preparedness, response, and cooperation (OPRC: Convention on Oil Pollution Preparedness, Response and Cooperation) was established. It facilitates international cooperation and mutual assistance among member states in the event of major pollution incidents [64].

By the late 2000s, a new type of convention emerged: the International Convention on Civil Liability for Bunker Oil Pollution Damage, also called the Bunker Convention. It succinctly establishes that, with few exceptions, the shipowner is liable for any pollution damage caused by bunker oil [65].

One of the oldest and most significant conventions is the International Oil Pollution Compensation Fund (FIPOL), created in 1971 and amended twice, in 1992 and 2003. This organization provides compensation for hydrocarbon pollution damages resulting from spills from oil tankers [66, 67].

Despite the establishment of these numerous conventions, regulation of offshore platforms and hydrocarbon transport faces several limitations. In general, the international conventions mentioned above are not specific enough to effectively regulate an activity as specialized as offshore oil exploitation. For example, an oil spill caused by a tanker grounding cannot be equated with one resulting from a platform accident [68].

The first limitation of these conventions is the international framework. While international cooperation can allow for rapid response, the transboundary nature of marine pollution often hampers intervention and fails to adequately protect the marine environment [48].

Additionally, the ocean represents an enormous portion of our planet—over 70% of the Earth's surface—with only 5% currently explored [69]. Visual observation of the marine environment is complex, and reproducing analogous conditions in laboratories to study underwater phenomena is difficult, if not nearly impossible. As a result, the marine environment still holds many mysteries that scientists attempt to unravel every year. Accurately and comprehensively assessing the risks posed by hydrocarbon exploitation to marine biodiversity is therefore impossible. With current knowledge, scientists cannot determine with complete certainty the full extent of damage industrial activities may cause to marine fauna and flora. Since we do not have perfect knowledge of these risks, regulations governing hydrocarbon exploitation are often not fully adequate [48].

Hydrocarbons also have significant economic value. Coupled with their role in geopolitical relations between states and in ensuring energy independence, hydrocarbons make the topic particularly sensitive, rendering oil activities difficult to regulate internationally [68].

Finally, due to the diversity in legal status of offshore platforms, each with complex and highly technical operations, legal fragmentation often complicates the understanding of the platform and the identification of applicable law [48].

Regulations That Can Be Implemented or Are Being Considered

European legislation focuses on persistent, bioaccumulative, and toxic contaminants (PBT), or very persistent and very bioaccumulative substances (vPvB), notably through the United Nations Stockholm Convention (Stockholm Convention, 2001). Very few polar compounds are regulated under this convention, and there is a regulatory gap for persistent, mobile, and toxic substances (PMT) [70]. The Stockholm Convention classifies these into two categories: PBT compounds, which are significant for internal exposure, and PMT compounds, which are significant for external exposure [27]. Polychlorinated biphenyls (PCBs), also called “pyralens,” accumulate in the environment and within food chains, posing a threat to ecosystems. They are notably used for their solvent resistance and low removal rate during forming processes [70].

Approximately 40,000 tons of oil were spilled following the Exxon Valdez grounding. This incident led to the rapid closure of commercial fisheries by the State of Alaska and caused widespread concern among indigenous communities as well as other seafood users. Although most

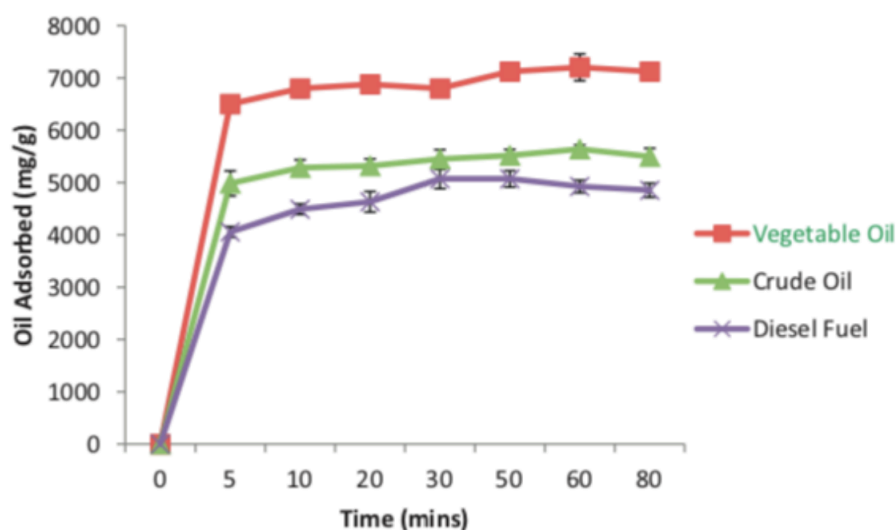


Figure 7: Absorption rate of African hair (60-70µm) in vegetal oil, crude oil and diesel fuel at 25°C. The lines represent the standardized mean. [72]

seafood products were found to be uncontaminated by polycyclic aromatic hydrocarbons (PAHs), local consumers were initially reluctant to accept these results [71]. Cod and shrimp fisheries were closed in the first days after the spill. In contrast, salmon and herring fishing continued throughout the rest of 1989. Indeed, shoreline cleanup activities relocated the oil to littoral zones where these species did not inhabit, allowing salmon and herring fisheries to continue. Furthermore, PAH concentrations were significantly higher in pink salmon tissues from oiled areas compared to unoiled areas, so fishing zones were more strictly monitored than before to prevent contamination [71].

By 1990, there was no longer any difference in PAH concentrations in tissues between oiled and unoiled areas, except in mussels, which still showed elevated contamination levels three to four years after the spill. The same phenomenon was observed in Lebanon in 2006 [24, 26].

Measures That Can Be Implemented by Local Associations

In addition to the use of chemical products such as dispersants, other methods are also employed to remove hydrocarbons spilled during accidents from seas and oceans. For example, floating barriers can prevent the horizontal spread of hydrocarbons, and mechanical recovery systems, also called skimmers, separate oily substances from water. Finally, the use of sorbents is an easy-to-implement and low-cost method for cleaning water by extracting hydrocarbons. Sorption is a physical process in which one substance is adsorbed or absorbed on or into another substance. A study was conducted to determine the sorption capacity of human hair for this purpose, testing three hair types: Asian, African, and European [72]. African hair absorbed the most hydrocarbons, followed by Asian and then European hair. Each hair type can absorb between three and nine times its weight in oil. The maximum absorption observed for African hair with crude oil was 8100 ± 165 mg/g [72].

Absorption occurs mainly in an exponential manner during the first five minutes, then slows down until reaching a plateau after 60 minutes (**Figure 7**), when all active sites of the hair are occupied. Moreover, hair can be reused multiple times without affecting its absorption capacity.

This makes it a cost-effective method, as efficient as synthetic, non-biodegradable materials previously used [72].

Regarding animals that have come into contact with hydrocarbons, several issues are observed, including inhalation of hydrocarbons (particularly PAHs), the hydrocarbon layer on the animal's body, and hydrocarbons in the digestive tract. Once ingested, hydrocarbons coat the digestive tract and prevent nutrient absorption. In a study published in March 2021 on the toxicity of oil to marine reptiles, menhaden oil or cod liver oil was mixed with mayonnaise, which also provides nutrition while removing the oil layer from the organism. This method is also used by the Turtle Hospital in Florida and recently at the National Sea Turtle Rescue Center in Israel. Because mayonnaise is composed of emulsified fats, it allows hydrocarbons to be captured and removed naturally while feeding the turtles. Additionally, if the animal is in very poor condition, a solution containing B vitamins, iron, calcium, and antibiotics can be administered. However, the long-term impacts of hydrocarbon ingestion remain unknown [18].

Conclusion

Oil spills have an undeniable harmful effect on the environment, but it is important to note that those caused by oil accidents represent only 10% of hydrocarbon releases, with the remaining 90% being chronic discharges, often due to individual activities [2]. However, an accident releases a massive amount of oil in high concentrations on a single biotope, posing far more problems than natural, random releases of smaller quantities [8].

Hydrocarbons, which constitute oil, have their own specific nomenclature [11]. Since their characteristics differ, their consequences are not always the same, which presents a challenge in developing management plans and regulations [48]. Moreover, beyond being an environmental risk, hydrocarbons can also be released naturally, emitting methane, which contributes to the greenhouse effect [4].

Beyond surface oiling, spills also impact the environment less visibly, affecting the water column, surface, depth, coastal zones, and even the atmosphere through volatile PAHs [12]. To mitigate their effects on biotopes, fauna, and flora, various methods have been developed to enhance biodegradation [27]. Often, hydrocarbons are not fully degraded naturally and remain suspended in the water column or deposited on the seafloor [30].

To accelerate the dispersion of oil slicks, dispersants have been used. Intended to fragment oil into droplets, dispersants were designed to make hydrocarbons more easily degradable [35]. However, they often have harmful effects on marine organisms and can sometimes cause even more severe consequences than the hydrocarbons themselves [12].

Although the development of less toxic dispersants has become a priority, hydrocarbons and their dispersants continue to affect cetaceans visibly [56]. Cetaceans can show severe immunodeficiency, lung diseases, or adrenal disorders, sometimes resembling conditions observed in humans [2].

References :

1. *Oil spills*. (2020). NOAA (National Oceanic and Atmospheric Administration). <https://www.noaa.gov/education/resource-collections/ocean-coasts/oil-spills>
2. Kuppusamy, S., Maddela, N. R., Megharaj, M., & Venkateswarlu, K. (2019). *Impact of total petroleum hydrocarbons on human health*. (pp. 139-165). Springer International Publishing.
3. Kolokoussis, P., & Karathanassi, V. (2018). *Oil spill detection and mapping using sentinel 2 imagery*. Journal of Marine Science and Engineering, 6(1).
4. Speight, J. (2007) *The chemistry and technology of petroleum, 4th edition*. CRC Press, Boca Raton.
5. Chalghmi, H. (2015). Etude de la pollution marine par les hydrocarbures et caractérisation de leurs effets biochimiques et moléculaires sur la palourde de Ruditapes sp (Doctoral dissertation, Université de Bordeaux).
6. Girin, M., Le Roux, A., & Rousseau, C. (2005). *Le sauvetage et la restauration du vivant dans les marées noires*. Bulletin de l'Académie vétérinaire de France.
7. Blaxter, J. H. S., & Ten Hallers-Tjabbes, C. C. (1992). The effect of pollutants on sensory systems and behaviour of aquatic animals. *Netherland Journal of Aquatic Ecology*, 26(1), 43-58.
8. National Research Council (2003). Studies, Division on Earth and Life, Board, M., Board, O. S., & Committee on *Oil in the Sea: Inputs, Fates, and Effects*. National academies Press.
9. Girin, M., & Mamaca, E. (2011). Mieux combattre les marées noires. Quæ.
10. Maes, F. (2004). National Research Council, Oil in the Sea III. Inputs, Fates and Effects, Washington, The National Academies Press. International journal of environment and pollution, 2003, 265 p.
11. Allaire-Verville, M. (2015). *Revue de la documentation scientifique sur la toxicité et les critères de qualités relatifs aux hydrocarbures pétroliers dans les sédiments*.
12. Milinkovitch, T., Kanan, R., Thomas-Guyon, H., & Le Floch, S. (2011). *Effects of dispersed oil exposure on the bioaccumulation of polycyclic aromatic hydrocarbons and the mortality of juvenile Liza ramada*. Science of the Total Environment, 409(9), 1643-1650.
13. Lehr, W., & Socolofsky, S. A. (2020). *The Importance of Understanding Fundamental Physics and Chemistry of Deep Oil Blowouts*. In *Deep Oil Spills* (pp. 14-24). Springer, Cham.
14. Journal Officiel de l'Union Européenne. (2013). Directive 2013/39/UE du Parlement Européen et du Conseil du 12 août 2013, modifiant les directives 2000/60/CE et 2008/105/CE en ce qui concerne les substances prioritaires pour la politique dans le domaine de l'Eau.

15. Lee, K., Boudreau, M., Bugden, J., Burridge, L., Cobanli, S. E., Courtenay, S., ... & Wohlgeschaffen, G. (2011). State of knowledge review of fate and effect of oil in the arctic marine environment. A report prepared for the National Energy Board of Canada.
16. Kuznetsov, A. N., Fedorov, Y. A., & Fattal, P. (2012), « *Les effets et le comportement du fioul déversé en 2007 dans le détroit de Kertch suite à un naufrage de pétrolier* », Méditerranée, 119 | 91-99.
17. Volkman, J. K., Holdsworth, D. G., Neill, G. P., & Bavor Jr, H. J. (1992). Identification of natural, anthropogenic and petroleum hydrocarbons in aquatic sediments. *Science of the Total Environment*, 112(2-3), 203-219.
18. Ruberg, E. J., Williams, T. D., & Elliott, J. E. (2021). Review of petroleum toxicity in marine reptiles. *Ecotoxicology*, 30(4), 525-536.
19. Liu, X., Chen, Z., Wu, J., Cui, Z., & Su, P. (2020). *Sedimentary polycyclic aromatic hydrocarbons (PAHs) along the mouth bar of the Yangtze River Estuary: Source, distribution, and potential toxicity*. *Marine Pollution Bulletin*, 159, 111494.
20. El Nemr, A., El-Sadaawy, M.M., Khaled, A. (2013). *Aliphatic and polycyclic aromatic hydrocarbons in the surface sediments of the Mediterranean: assessment and source recognition of petroleum hydrocarbons*. *Environ Monit Assess* 185, 4571–4589.
21. Lerda, D. (2011). *Polycyclic aromatic hydrocarbons (PAHs) factsheet*. Belgium, Joint Research Centre, European Commission, 34.
22. He, Y., He, W., Yang, C., Liu, W., & Xu, F. (2020). *Spatiotemporal toxicity assessment of suspended particulate matter (SPM)–bound polycyclic aromatic hydrocarbons (PAHs) in lake chaohu, china: Application of a source-based quantitative method*. *The Science of the Total Environment*, 727, 138690-138690. <https://doi.org/10.1016/j.scitotenv.2020.138690>
23. Milinkovitch, T. (2011). *Stratégie de lutte contre les catastrophes pétrolières et risque environnemental associé: évaluation de la toxicité d'un dispersant en milieu côtier chez Liza sp* (Doctoral dissertation, université de La Rochelle), 24-37.
24. Azoury, S. (2013). *Étude de la contamination de la côte libanaise par les hydrocarbures aromatiques polycycliques (HAPs) et les éléments traces métalliques (ETMs): archives sédimentaires et biomonitoring suite à une marée noire* (Doctoral dissertation, Université Sciences et Technologies-Bordeaux I).
25. Deyme, R., Bouloubassi, I., Taphanel-Valt, M., Miquel, J., Lorre, A., Marty, J., & Méjanelle, L. (2011). *Vertical fluxes of aromatic and aliphatic hydrocarbons in the northwestern mediterranean sea*. *Environmental Pollution* (1987), 159(12), 3681-3691.
26. Law, R. J., & Hellou, J. (1999). *Contamination of fish and shellfish following oil spill incidents*. *Environmental Geosciences*, 6(2), 90-98.

27. Kristensen, M., Johnsen, A. R., & Christensen, J. H. (2021). *Super-complex mixtures of aliphatic-and aromatic acids may be common degradation products after marine oil spills: A lab-study of microbial oil degradation in a warm, pre-exposed marine environment*. Environmental Pollution, 117264.
28. De Oliveira Soares, M., Teixeira, C. E. P., Bezerra, L. E. A., Paiva, S. V., Tavares, T. C. L., Garcia, T. M., & Cavalcante, R. M. (2020). *Oil spill in South Atlantic (Brazil): Environmental and governmental disaster*. Marine Policy, 115, 103879.
29. Mahajan, M. C., Phale, P. S., & Vaidyanathan, C. S. (1994). *Evidence for the involvement of multiple pathways in the biodegradation of 1-and 2-methylnaphthalene by Pseudomonas putida CSV86*. Archives of microbiology, 161(5), 425-433.
30. Doumenq, P., Guiliano, M., de Beaulieu, J. L., & Bertrand, J. C. (2005). *Origines, devenir et effets des hydrocarbures dans les sédiments marins*. Natures Sciences Sociétés, 13(1), 76-80.
31. Gilbert, F., Stora, G., Desrosiers, G., Deflandre, B., Bertrand, J., Poggiale, J., & Gagné, J. (2001). Alteration and release of aliphatic compounds by the polychaete nereis virens (sars) experimentally fed with hydrocarbons. *Journal of Experimental Marine Biology and Ecology*, 256(2), 199-213.
32. Serrano, A., Tejada, M., Gallego, M., & Gonzalez, J. L. (2009). Evaluation of soil biological activity after a diesel fuel spill. *The Science of the Total Environment*, 407(13), 4056-4061. <https://doi.org/10.1016/j.scitotenv.2009.03.017>
33. Chapman, H., Purnell, K., Law, R. J., & Kirby, M. F. (2007). The use of chemical dispersants to combat oil spills at sea : A review of practice and research needs in Europe. *Marine Pollution Bulletin*, 54(7), 827-838.
34. Almeda, R., Hyatt, C., & Buskey, E. J. (2014). *Toxicity of dispersant Corexit 9500A and crude oil to marine microzooplankton*. Ecotoxicology and environmental safety, 106, 76-85.
35. Sintef, P. D. & A. Lewis (2001). "Oil Spill Dispersants." 27.
36. Alo, C., Dallongeville, A., Youssouf, A. M., & Vin, B. (2011). Evaluation des risques chroniques liés à l'exposition aux polluants chimiques des primo-intervenants lors de la collecte de pétrole brut lourd déversé en mer avec ou sans l'utilisation de dispersants, 57-90.
37. Cohen, J. H., McCormick, L. R., & Burkhardt, S. M. (2014). *Effects of dispersant and oil on survival and swimming activity in a marine copepod*. Bulletin of environmental contamination and toxicology, 92(4), 381-387.
38. US-EPA (2006). Exposure Factors Handbook, Dermal Exposure Factors, Office of Research and Development National Center for Environmental Assessment, US-EPA.
39. Van Der Ham, J. L., & De Mutsert, K. (2014). *Abundance and size of gulf shrimp in Louisiana's coastal estuaries following the deepwater horizon oil spill*. PLoS One, 9(10), e108884.

40. Duke, N. C., Burns, K. A., Swannell, R. P., Dalhaus, O., & Rupp, R. J. (2000). Dispersant use and a Bioremediation Strategy as alternate means of reducing impacts of large oil spills on mangroves : the gladstone field trials. *Marine Pollution Bulletin*, 41(12), 403-412.
41. Baca, B., Ward, G. A., Lane, C. H., & Schuler, P. A. (2005). Net environmental benefit analysis (NEBA) of dispersed oil on nearshore tropical ecosystems derived from the 20 year "TROPICS" field study. *International Oil Spill Conference Proceedings*, 2005(1), 453-456.
42. Ramachandran, S., Hodson, P. V., & Lee, K. (2003). *Significance of oil droplets in chemically enhanced water-accommodated fraction*.
43. Lin, C. Y., Anderson, B. S., Phillips, B. M., Peng, A. C., Clark, S., Voorhees, J., Wu, H. D. I., Martin, M. J., McCall, J., Todd, C. R., Hsieh, F., Crane, D., Viant, M. R., Sowby, M. L., & Tjeerdema, R. S. (2009). Characterization of the metabolic actions of crude versus dispersed oil in salmon smolts via NMR-based metabolomics. *Aquatic Toxicology*, 95(3), 230-238.
44. Gulec, I., Leonard, B., & Holdway, D. A. (1997). Oil and dispersed oil toxicity to amphipods and snails. *Spill Science & Technology Bulletin*, 4(1), 1-6.
45. Long, S., & Holdway, D. A. (2002). Acute toxicity of crude and dispersed oil to *Octopus pallidus* (Hoyle, 1885) hatchlings. *Water Research*, 36(11), 2769-2776.
46. Wells, P., & Keizer, P. (1975). Effectiveness and toxicity of an oil dispersant in large outdoor salt water tanks. *Marine Pollution Bulletin*, 6(10), 153-157.
47. Thorhaug, A., Marcus, J., & Booker, F. (1986). Oil and dispersed oil on subtropical and tropical seagrasses in laboratory studies. *Marine Pollution Bulletin*, 17(8), 357-361.
48. Truilhé, E., & Bouillard, C. (2017). *Quel encadrement juridique pour les activités pétrolières offshore en droit de l'Union européenne?*.
49. Kloff, S., & Wicks, C. (2005). *Gestion environnementale de l'exploitation de pétrole offshore et du transport maritime pétrolier*. Fondation internationale du Banc d'Arguin.
50. Loudy, I., (2015) *Stratégies discursives utilisées par les entreprises en réponse aux catastrophes environnementales en vue de réparer leur légitimité : le cas de l'industrie pétrolière*.
51. Antonio, F. J., Mendes, R. S., & Thomaz, S. M. (2011). *Identifying and modeling patterns of tetrapod vertebrate mortality rates in the Gulf of Mexico oil spill*. *Aquatic Toxicology*, 105(1-2), 177-179.
52. Alloy, M. M., Boubé, I., Griffitt, R. J., Oris, J. T., & Roberts, A. P. (2015). *Photo-induced toxicity of Deepwater Horizon slick oil to blue crab (Callinectes sapidus) larvae*. *Environmental toxicology and chemistry*, 34(9), 2061-2066.
53. Beyer, J., Trannum, H. C., Bakke, T., Hodson, P. V., & Collier, T. K. (2016). *Environmental effects of the Deepwater Horizon oil spill: a review*. *Marine pollution bulletin*, 110(1), 28-51.

54. Brette, F., Machado, B., Cros, C., Incardona, J.P., Scholz, N.L., Block, B.A., (2014). *Crude oil impairs cardiac excitation–contraction coupling in fish*. Science 343, 772–776.
55. Jung, J. H., Yim, U. H., Han, G. M., & Shim, W. J. (2009). Biochemical changes in rockfish, *Sebastes schlegeli*, exposed to dispersed crude oil. Comparative Biochemistry and Physiology Part C : Toxicology & Pharmacology, 150(2), 218-223.
56. Carmichael RH, Graham WM, Aven A, Worthy G, Howden S (2012) *Were multiple stressors a 'perfect storm' for northern Gulf of Mexico bottlenose dolphins (Tursiops truncatus) in 2011?*. PLoS One 7:e41155
57. Wise, C.F., Wise, J.T.F., Wise, S.S., Thompson, W.D., Wise Jr., J.P., Wise Sr., J.P., (2014). *Chemical dispersants used in the Gulf of Mexico oil crisis are cytotoxic and genotoxic to sperm whale skin cells*. Aquat. Toxicol. 152, 335–340.
58. DeLeo, D. M., Herrera, S., Lengyel, S. D., Quattrini, A. M., Kulathinal, R. J., & Cordes, E. E. (2018). *Gene expression profiling reveals deep-sea coral response to the Deepwater Horizon oil spill*. Molecular ecology, 27(20), 4066-4077.
59. Hsing, P. Y., Fu, B., Larcom, E. A., Berlet, S. P., Shank, T. M., Govindarajan, A. F., & Thomsen, L. (2013). *Evidence of lasting impact of the Deepwater Horizon oil spill on a deep Gulf of Mexico coral community Oil spill impact on deep coral community*. Elementa: Science of the Anthropocene, 1.
60. Epstein, N., Bak, R. P. M. & Rinkevich, B. (2000). Toxicity of third generation dispersants and dispersed egyptian crude oil on red sea coral larvae. Marine Pollution Bulletin, 40(6), 497-503.
61. Hu, H., Zhang, J., & Chen, W. (2011). *Competition of bloom-forming marine phytoplankton at low nutrient concentrations*. Journal of Environmental Sciences, 23(4), 656-663.
62. Lee, L. T. (1983). *The law of the sea convention and third states*. Am. J. Int'l L., 77, 541..
63. Peet, G. (1992). *The MARPOL Convention: implementation and effectiveness*. International Journal of Estuarine and Coastal Law, 7(4), 277-295).
64. Kim, K. S. (2006). *The present state of domestic acceptance of various international conventions for the prevention of marine pollution*. Journal of the Korean Society of Marine Environment & Safety, 12(4), 293-300.
65. Zhu, L. (2007). *International Convention on Civil Liability for Bunker Oil Pollution Damage, 2001—Liability and Insurance Aspects*. In Pollution of the Sea—Prevention and Compensation (pp. 171-180). Springer, Berlin, Heidelberg)
66. Hay, J. (2006). *Analyse économique du système international CLC/FIOPOL comme instrument de prévention des marées noires* (Doctoral dissertation, Université de Bretagne occidentale-Brest).
67. Jacobsson, M. (1996). *Oil pollution liability and compensation: an international regime*. Uniform Law Review-Revue de droit uniforme, 1(2), 260-272.

68. Rochette, J., Chabason, L., Callet, S., & Wemaere, M. (2014). *To put an end to the oil blue. For a better regulation of offshore oil and gas activities.*
69. Caron, J. M., & Alain, G. (2007). *Planete terre*. Editions OPHRYS.
70. Reemtsma, T., Berger, U., Arp, H. P. H., Gallard, H., Knepper, T. P., Neumann, M., ... & Voogt, P. D. (2016). Mind the Gap: Persistent and Mobile Organic Compounds. Water Contaminants That Slip Through.
71. Law, R. J., & Hellou, J. (1999). Contamination of fish and shellfish following oil spill incidents. *Environmental Geosciences*, 6(2), 90-98.
72. Ifelebuegu, A. O., Anh Nguyen, T. V., Ukotije-Ikwut, P., & Momoh, Z. (2015). *Liquid-phase sorption characteristics of human hair as a natural oil spill sorbent*. *Journal of Environmental Chemical Engineering*, 3(2), 938-943.